

Literature Review Detailed: Key Results and Intuition

Draft notes

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This note records, paper by paper, the key results and the intuition behind them. It is more detailed than the positioning note (`literature_view`).

1 Public finance and deadweight loss

1.1 The Measurement of Waste (Harberger, 1964)

Harberger (1964) develops a linearized framework for measuring aggregate efficiency losses from taxes and related price distortions. The paper emphasizes six results.

- **Measuring deadweight loss (ΔW) as a policy tool**

- **Key result:** Deadweight loss can be calculated with a linearized aggregate framework that measures efficiency changes across distorted markets:

$$\Delta W = \frac{1}{2} \sum_i \sum_j R_{ij} T_i T_j.$$

- **Intuition:** Rather than only asking whether an allocation is Pareto optimal, policy-makers need a quantitative measure of efficiency loss to weigh tax reforms, tariffs, and other real-world trade-offs.

- **Neutrality of uniform value-added taxes**

- **Key result:** An equal percentage tax on value added across all productive activities yields zero net efficiency loss:

$$\sum_j v_j R_{ij} = 0 \implies \Delta W = 0 \quad \text{when } t_i = t \text{ for all } i.$$

- **Intuition:** Uniform value-added taxation leaves relative price ratios across activities unchanged, so it does not induce distortionary factor substitution.

- **Equivalence of product taxes and value-added taxes**

- **Key result:** Product taxes can be translated into equivalent taxes on value added when intermediate inputs enter output in fixed proportions.
- **Intuition:** Taxing final output creates the same behavioral incentives and equilibrium as taxing all factor inputs (labor, capital, materials) at the same effective rate.

- **Cross-tax distortions and welfare-improving tax increases**

- **Key result:** Introducing or raising a tax on a good can raise total social welfare if that good is a substitute for another already heavily taxed good:

$$\frac{\partial(\Delta W)}{\partial T_k} > 0 \quad \text{when } T_j > 0 \text{ and } R_{kj} > 0.$$

- **Intuition:** If Good A is heavily taxed, consumers substitute toward Good B . Taxing Good B shifts demand back toward Good A , reducing the preexisting distortion in Sector A by enough to offset the new distortion in Sector B .

- **Distortion of the labor–leisure choice**

- **Key result:** A uniform tax on all market labor is non-neutral unless leisure (non-market activity) can also be taxed:

$$\sum_{j=1}^{n+1} M_{ji} = 0.$$

- **Intuition:** Leisure is hard to tax. Taxing all market labor therefore induces substitution into untaxed leisure and creates an efficiency loss in labor supply.

- **Capital–labor cross-factor interaction effects**

- **Key result:** When taxes distort both capital (B_i) and labor (E_i), cross-factor interaction terms ($H_{ij} = \partial K_i / \partial E_j$) must be included to measure total efficiency loss:

$$\Delta W = \frac{1}{2} \sum_i \sum_j M_{ij} E_i E_j + \frac{1}{2} \sum_i \sum_j G_{ij} B_i B_j + \sum_i \sum_j H_{ij} B_i E_j.$$

- **Intuition:** A tax on capital in one industry reallocates both capital and labor across sectors. Ignoring how labor responds to capital taxation (and vice versa) misstates total efficiency loss.

References

Arnold C. Harberger. The Measurement of Waste. *The American Economic Review*, 54(3):58–76, 1964. ISSN 0002-8282. URL <https://www.jstor.org/stable/1818490>.